

Introduction. Examples and alternative approaches

G0B75A Meta-analysis

Prof. Wim Van den Noortgate

Outline

- Two examples
- Alternative approaches

An example in education

(Raudenbush, S. W. (1984). Magnitude of teacher expectancy effects on pupil IQ as a function of the credibility of expectancy induction: A synthesis of findings from 18 experiments. *Journal of Educational Psychology*, 76, 85-97.)

Study	Weeks prior contact	g_j	$\sigma_{(g_j)}$
1. Rosenthal et al. (1974)	2	0.03	0.13
2. Conn et al. (1968)	3	0.12	0.15
3. Jose & Cody (1971)	3	-0.14	0.17
4. Pellegrini & Hicks (1972)	0	1.18	0.37
5. Pellegrini & Hicks (1972)	0	0.26	0.37
6. Evans & Rosenthal (1969)	3	-0.06	0.10
7. Fielder et al. (1971)	3	-0.02	0.10
8. Claiborn (1969)	3	-0.32	0.22
9. Kester & Letchworth (1972)	0	0.27	0.16
10. Maxwell (1970)	1	0.80	0.25
11. Carter (1970)	0	0.54	0.30
12. Flowers (1966)	0	0.18	0.22
13. Keshock (1970)	1	-0.02	0.29
14. Henrickson (1970)	2	0.23	0.29
15. Fine (1972)	3	-0.18	0.16
16. Greiger (1970)	3	-0.06	0.17
17. Rosenthal & Jacobson	1	0.30	0.14
18. (1968)	2	0.07	0.09
19. Fleming & Anttonen (1971)	3	-0.07	0.17

Conclusions:

- Mean effect: 0.084, $p = .11$
- Initial effect is substantial (0.41, $p < .001$), but decreases with increasing prior contact (with -0.16 per week, $p < .001$)

An example in medical research

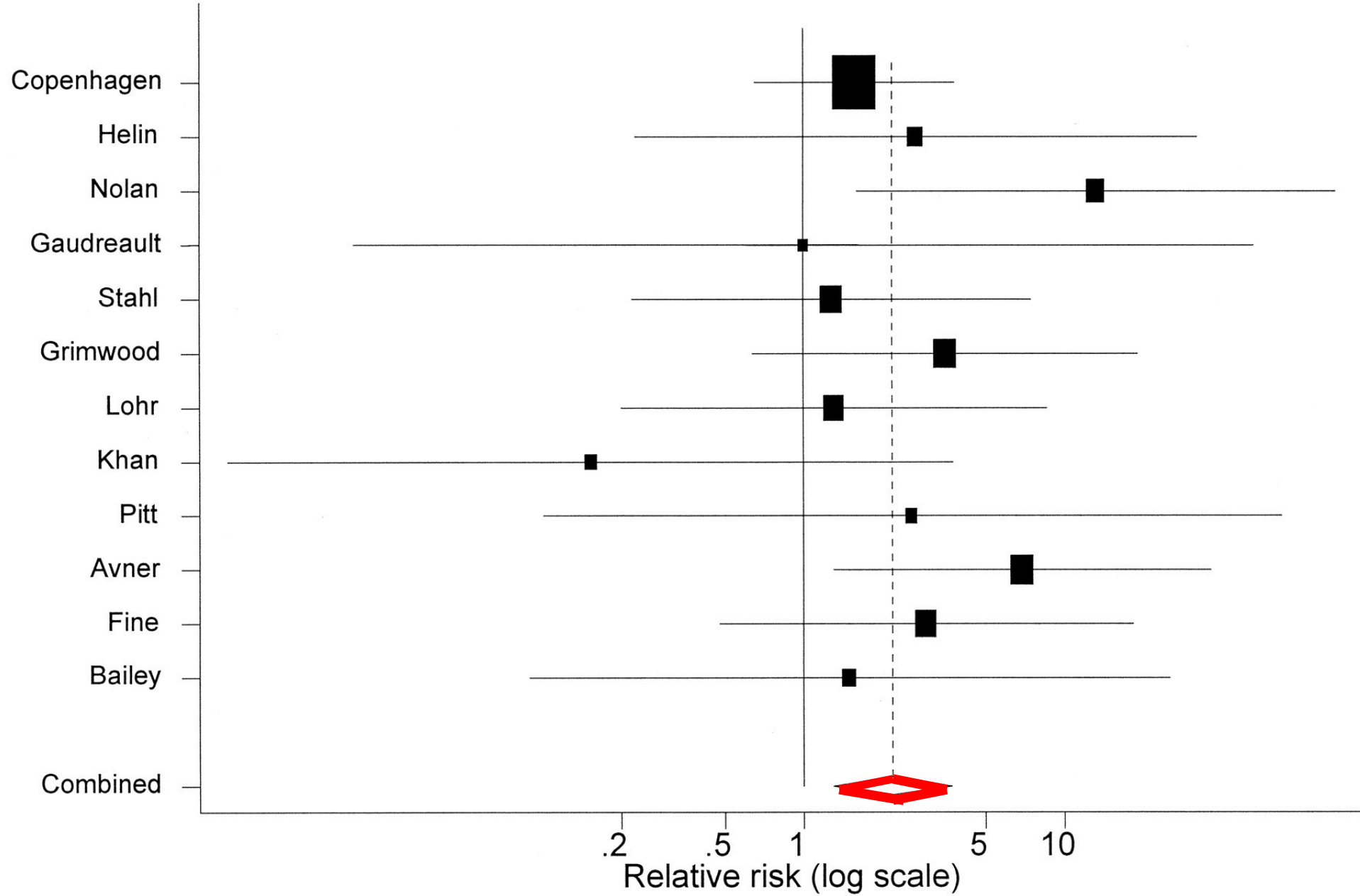
(Keren, R., & Chan, E. (2002). A meta-analysis of randomized, controlled trials comparing short- and long-course antibiotic therapy for urinary tract infections in children. *Pediatrics*, 109, e70.)

Study	Year	Sample Size	RR (95% CI)
Bailey and Abbott	1978	10	1.33 (0.17–10.25)
Khan et al	1981	16	0.20 (0.01–3.61)
Stahl et al	1984	26	1.20 (0.34–4.28)
Fine and Jacobson	1985	31	2.34 (0.53–10.30)
Gaudreault et al	1992	40	1.00 (0.02–48.09)
Pitt et al	1982	42	2.50 (0.11–58.06)
Helin	1984	43	2.53 (0.25–25.81)
Grimwood et al	1988	45	2.80 (0.65–12.02)
Auman et al	1992	49	4.60 (1.12–18.61)

- RR:

RR = Relative Risk = Risk Ratio

$$= \frac{\pi_{\text{treatment failure in short course antibiotic treatment}}}{\pi_{\text{treatment failure in long course antibiotic treatment}}}$$



Combined RR = 1.94 (95% CI: 1.19–3.15)

Conclusions:

“In all but 2 of the previous RCTs of short- versus long-course therapy for UTI in children, there was no statistically significant difference in treatment failure rates. Most authors interpreted this **failure to achieve statistical significance** as evidence that short-course therapy was at least as effective as long-course therapy. An alternative explanation, however, as noted by Shapiro in a 1982 review, is that there was a difference in efficacy but that these studies, because of small sample size, simply did **not have the power** to detect this difference.”

“When we pooled the results of these nonsignificant studies, we gained sufficient power to show that short-course therapy was indeed associated with a **statistically significant 94% increase** in treatment failures compared with long-course therapy.”

Outline

- Two examples
- Alternative approaches

Vote counting procedures

- 3 categories: negative effect, no effect, positive effect
- Category with most 'votes' or the largest proportion of studies 'wins'

Discussion of vote-counting procedures

- Simple and intuitive

- ☞ Low power

- ☞ Power does not necessarily increase with an increasing number of studies

- ☞ All studies are given the same weight

- ☞ Little information obtained about
 - the size of the effect
 - the homogeneity of the effect
 - when the effect occurs

Combining p -values

E.g., Stouffer method:

- Under H_{0i} in primary study i : $p_i \sim U(0,1)$

$$\Rightarrow z(p_i) \sim N(0,1)$$

- If all H_0 in a set of independent studies are true:

$$\Rightarrow \frac{\sum_{i=1}^k z(p_i)}{\sqrt{k}} \sim N(0,1)$$

Discussion of combining p -values

- Applicable even if studies differ from each other in design, in tests that are used
- Nonparametric test (allows also nonparametric tests in primary studies)
- ☞ Interpretation: alternative hypothesis: at least one of the H_0 is not correct
- ☞ Little information obtained about
 - the size of the effect
 - the homogeneity of the effect
 - When the effect occurs

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